Optimized Deep Learning Architectures Using EfficientNet for Diabetic Retinopathy Diagnosis

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2025 World Skills Conference on Universal Data Analytics and Sciences (WorldSUAS) | 979-8-3315-3925-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/WorldSUAS66815.2025.11199165

***Abstract*—Diabetic retinopathy (DR) is a leading cause of vision loss in diabetic patients. Early detection and classification are essential for effective treatment. DR can damage vision and may even lead to blindness if it is not diagnosed in the early stages. The expert analyzes colored fundus images to diagnose DR. However, this manual process is time-consuming and prone to errors. Computer vision techniques aid in the automated detection of DR and its stages from retinal images. This study focuses on improving DR severity detection and classification using advanced models like transfer learning. The research goal is to enhance diagnostic accuracy using the EfficientNet model to reduce the diagnosis time and determine the disease's severity levels. However, previous existing methods struggle to capture complex features and classify DR stages accurately, especially in the early stages. In this research, we used the publicly available Kaggle dataset of retina images to train the deep convolutional neural network (CNN) model, EfficientNet, to capture detailed features, improve the DR stage, and achieve good accuracy by using Adam optimizers. This research highlights that the deep learning (DL) experimental results show that the proposed model accurately detects all DR stages, unlike the current methods, and performs better compared to state-of-the-art methods on the same Kaggle dataset**

Keywords—Deep learning, efficientNet, diabetic retinopathy, retinal fundus images, artificial intelligence, APTOS dataset 2019.

I. Introduction

Diabetes has quickly grown into one of the most widespread health issues in the last few decades. Diabetic retinopathy (DR) often leads to blindness in many people [1]. Increased rates of diabetes around the world are leading to a higher number of DR cases. The risk of DR is higher for those who have had diabetes for a significant amount of time. A complication resulting from diabetes may be that the vessels in the retina swell and start leaking blood and fluid in the posterior part of the eye[2]. Its severity is used to divide DR into five different grades. The different stages are 1) no DR, 2) mild DR, 3) moderate DR, 4) severe DR, and 5) proliferative DR. Only the first one, non-proliferative retinopathy, does not have DR. Mild, moderate, and severe form a part of the non-proliferative DR called NPDR (Non-Proliferative DR)[3]. In the NPDR stage, patients typically do not experience noticeable symptoms. The only way to detect NPDR is through a fundus examination conducted by a trained ophthalmologist. If left untreated, the condition can progress to the Proliferative DR (PDR)[4]. At times, abnormal blood vessels develop and appear at the back of the eye around this phase. A burst in these vessels can result in blood leaking in the eye, causing things to look blurry and eventually resulting in blindness. It is best to treat diabetes retinopathy in its early stages, which is before it gets worse. For this reason, inspecting the retinas of diabetic patients on a regular basis helps catch DR early. Being diagnosed and treated on time is important for diabetics to avoid losing their sight [5]. The different phases of DR can be seen in Fig. 1. By looking at the image, it is clear that there is little difference between the Normal and Mild stages. It can be difficult to identify the Mild stage as it looks much like the normal stage.

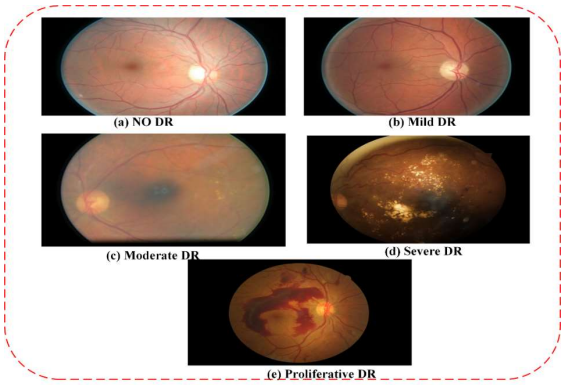

Fig. 1. The different stages of DR

The number of diabetics in India will surpass 69.9 million by 2025 and will continue to grow to 80 million by 2030. According to the International Diabetes Federation (IDF)-10th Edition, in 2021, one in ten adults (537 million) aged 20–79 had diabetes. This number is expected to rise to 643 million by 2030, 783 million by 2045 [6], and 1.3 billion by 2050, highlighting the rapid global increase in diabetes cases. The initial stage of DR shows no symptoms in patients before its advanced development starts, causing edema together with blurred vision and visual distortions, and visual acuity. Early detection of DR remains challenging yet essential because it requires supreme effort to identify this condition during its initial phases. Preventive measures should be taken to stop advanced-stage complications from developing.

The number of diagnostic systems powered by DL assistance has increased during the last decade while surpassing older image processing-based systems. The medical field widely uses this technology for its ability to detect cancerous tumors in both lungs and breasts while diagnosing COVID-19 from CT scans. The quick advancements in DL-based computer vision technology resulted in the development of advanced neural network structures that significantly upgraded model performance levels[7].

The significant contribution of this research work is as follows:

- We utilized a state-of-the-art EfficientNet model, which enabled us to detect blindness symptoms at their earliest instances, leading to enhanced blindness identification from retinal images with a validation accuracy exceeding 96%, compared to CNN models.
- Our study compares various transfer learning models, including machine learning, deep learning,

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and our proposed model, to identify the best for DR diagnosis through detailed performance analysis using multiple evaluation metrics.

- The study used image preprocessing techniques like rotation, flipping, resampling, filling, and zooming to improve retinal fundus images by removing artifacts and noise.

The rest of the article is structured in the following sections: Section 2, contains a state-of-the-art literature review. In Section 3, discusses the methodology as well as the method used. In Section 4, details about the experiments are given, and a discussion and evaluation of the results. In Section 5, the article conclusion.

II. Literature Review

In the past, several research works focused on retinal disease detection from fundus images using machine learning algorithms to classify large amounts of fundus images obtained from the retinal screening program. In this context, DL techniques are emerging and being utilized for the early detection of DR and preventing vision loss, which will enhance the state of global health. In paper[8], HDR-EfficientNet is a DL model for classifying hypertensive and DR using retinal fundus images. It leverages EfficientNet-V2 with spatial-channel attention and transfer learning to improve accuracy and generalization while addressing class imbalance. Trained on 36,000 augmented images, it achieves high diagnostic performance. However, its limitations include limited dataset size, class imbalance issues, and high computational requirements, which may impact real-world applicability.

In study[9], the author utilized while AI and machine learning models like Linear SVC and Logistic Regression show strong potential in DR prediction, challenges such as limited interpretability, transparency, and slow clinical adoption persist. Key predictive factors include HbA1c levels and disease duration. The study underscores the value of data mining and the growing importance of explainable AI (XAI) in enhancing diagnostic accuracy and clinician trust. It also highlights the need to overcome traditional diagnostic biases and improve current methods for more effective DR screening.

In study[10], the author combines improved DR classification using SLIC-G and explores neural networks like GANs and CNNs for fundus image analysis. It emphasizes image processing, AI-based treatment approaches (e.g., Federated and Deep Learning), and early detection. The proposed method introduces a Dynamic Routing-CapsNet (DR-CN) integrating Capsule Networks and EfficientNet for feature extraction, with pre-processing techniques enhancing performance. The model effectively captures feature relationships, achieving high accuracy in early DR diagnosis. In paper[11], the author presents an accurate and efficient model for automated DR detection by integrating EfficientNet with Dynamic Routing-CapsNet. This combination enhances classification performance and supports early diagnosis. The approach shows promise in improving patient outcomes, though further research is needed for broader clinical adoption.

In study[12], the author examines benchmark CNNs like Inception-V3, GoogLeNet, and VGG-Net for DR classification, noting the lack of unified evaluation on the APTOS dataset. The study proposes a comparative analysis using CNNs with LSTM and GRU, transfer learning, and data augmentation to enhance classification accuracy and robustness. In paper[13] explores various DR detection techniques, including a reformed capsule network validated on the Messidor dataset for multi-stage DR classification. It highlights the use of DenseNet on the APTOS dataset with 96.11% accuracy and a segment-based learning approach to improve accuracy. EfficientNet-B3 also demonstrated high performance, achieving a kappa score of in DR severity assessment. The review emphasizes the role of transfer learning and pre-trained models in enhancing classifier effectiveness for accurate DR detection.

III. Methods and Methodologies

This research presents findings addressing a DL model leveraging transfer learning techniques to precisely diagnose DR. The study employed EfficientNet, a state-of-the-art convolutional neural network (CNN), pretrained on retinal images. As shown in Fig. 2, the proposed methodology outlines a step-by-step process and framework of the developed solution. Specifically, EfficientNet was fine-tuned using retinal scans annotated with medical classifications. Next, the adapted model was extensively tested against unseen patient data to confirm its ability to accurately detect various stages of DR. Overall, the presented transfer learning approach demonstrates promise for streamlining the diagnosis and treatment of this vision-endangering condition.

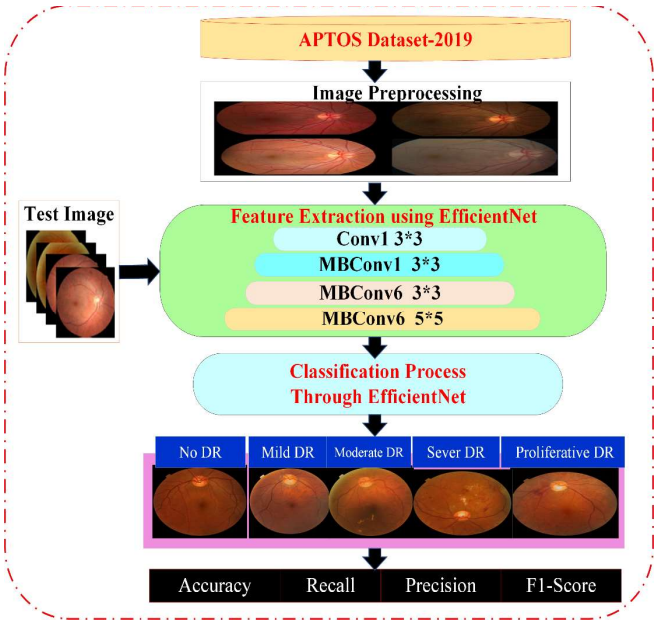

Fig. 2. The proposed Resnet50 transfer learning model for multiclass DR classification

A. Dataset

It is important to pick out a diverse set of images when selecting the dataset. For this project, we used open datasets of fundus images that came with annotations of the image severity. With enough images, the dataset APTOS (Asia Pacific Ophthalmology Society-2019) allows for good results in deep-CNN learning. Kaggle APTOS contains a total of 3672 images, spread out over 5 stages of DR severity, with 2930 in the training set, 366 in the validation set, and 366 in the test set [12]. The retinopathy level at the image level is given as a DR grade from 0 to 4, showing the presence of no

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retinopathy or proliferative retinopathy. The collection has 3672 images, all of which are organized into five groups depending on the DR stage. The categories are No DR (0), Mild DR (1), Moderate DR (2), Severe DR (3), and Proliferative DR (4). It covers No DR with 1,815 images, Mild DR with 370, Moderate DR with 999, Severe DR with 193, and Proliferative DR with 295. Thanks to this system, researchers and doctors can analyze and detect DR since it captures its stages of progression. Table I proves that there are representative images for every one of the five severity levels found in the dataset.

TABLE I. DATASETS AND ITS SAMPLE IMAGES

Dataset/ Class	No DR	Mild	Moderate	Severe	Proliferative
APTOS Dataset	1815	370	999	193	295

B. Data Pre-processing

In the proposed EfficientNet model, input images are resized to 224x224x3 pixels before preprocessing so that they will be at the correct size required when pre-processed. A variety of data augmentations are used to increase the size of the training dataset. This includes adding various modifications to these images[14]. We may need to adjust, such as changes in height and width, shear transformation, rotation, or zoom, turning about one axis with a given direction, and turning about two axes around a point beyond the object centre, or reflections of brightness compared to the dark image. Other modifications include vertical and horizontal flipping, and brightness enhancement. At regular intervals, we shall utilize the method of modifying hyperparameters for it to adapt its control system However, these improvements serve only a minor. For the accuracy and reliability of our data sets, we will use a Python installation. It is called an "Image Data Generator" program that will prepare large volumes of consistent images in batches. The original un-augmented data should still be used for testing and validation because there is no need to augment it. A data frame named "df" has been created to store the image paths along with their corresponding labels. Figure 3 shows the sample photos that were provided following preprocessing.

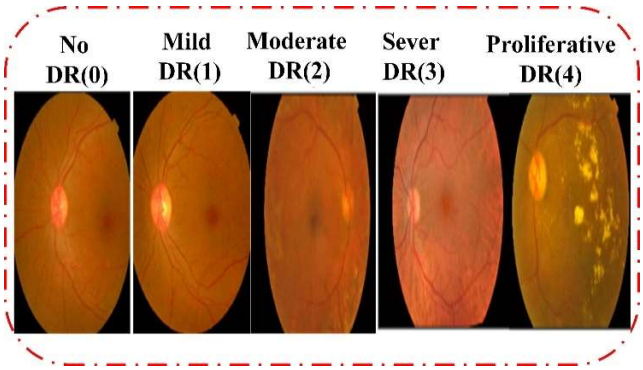

Fig. 3. Pre-processed images for DR severity categories

It offers batches of enhanced images as part of the data generator. The training parameter will be used to identify whether the data is being utilized for training. Otherwise, it used valid_test_datagen for testing and validation. Three datasets—the valid dataset, the train dataset, and the test dataset—are ultimately produced following the configuration of the data generators. The next batch of data, which consists of images and their labels, is then retrieved once the classes are extracted using their indices. Twenty plots, each showing a picture that corresponds to its DR classification, will be shown using a grid. There are five classes in the DR classification, with numbers 0 through 4.

C. Transfer Learning Model

The EfficientNet is a well-established architecture in transfer learning models. It presented a new model scaling technique that uses a compound coefficient to scale all depth, breadth, and resolution dimensions evenly. EfficientNet models have been widely adopted due to their strong performance and efficiency in terms of accuracy vs. computational cost[15]. EfficientNet is comprised of seven stages, and this configuration is replicated a specified number of times (refer to Fig. 2). The EfficientNet model's capacity to learn global features that are specific to the dataset. The comprehensive framework of the proposed model is presented in Fig. 4.

The fundamental idea behind this architecture is to scale neural networks in a way that maximizes their breadth, height, and depth. A shallow classifier that classifies five different class labels, which correspond to the DR grades, is used in place of the fully connected layer (FC1000 layer), while the convolutional layers' parameters remain unchanged. Achieving an ideal balance in neural network scaling with respect to width, height, and depth is the central idea of this architecture. Our recommendation is to use the EfficientNet version in order to strike a balance between running duration and performance.

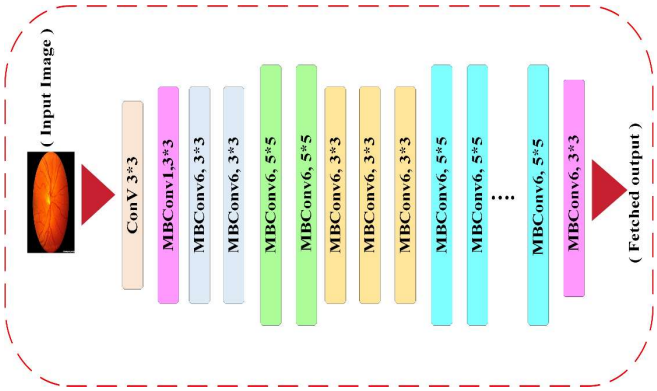

Fig. 4. Fundamental blocks of EfficientNet architecture

IV. RESULTS AND DISCUSSIONS

The investigation was executed using Google Colab on a laptop having an Intel Core i5 7200U processor, a dedicated GeForce 940 MX graphics processing unit (GPU), and 16GB of RAM, providing sufficient computational capacity for the tasks at hand. This arrangement provided the environment in which the proposed transfer learning model could be trained and tested. Performance metrics, derived from equations 1 to 4, were utilized to estimate the effectiveness of the model. These metrics suggest valuable insights interested in the accuracy, precision, recall, and overall effectiveness of the proposed approach. The computational setup and metrics highlight the reliability of the proposed approach.

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$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \quad (1)$$

$$Precision = \frac{Trpos}{Trpos + Flspos} \quad (2)$$

$$Recall = \frac{Trpos}{Trpos + Flsneg} \quad (3)$$

$$F1 - Score = 2 * \frac{Recall \times Precision}{Recall + Precision} \quad (4)$$

The dataset is generated by combining two sources from Kaggle: the DR-classification APTOS dataset. The dataset comprises five classes: Type 1: No DR; Type 2: Mild DR; Type 3: Moderate DR; Type 4: Severe DR; Type 5: Proliferative DR. The quantity of sample photos available in each class for testing and training is shown in Fig. 5.

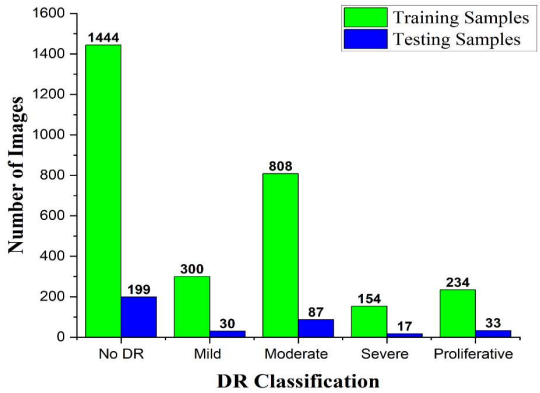

Fig. 5. The data samples of each class of DR

Figure 6 presents the accuracy charts for both training and validation phases. Our proposed model demonstrates strong performance, achieving 98% accuracy during training and 96.14% accuracy when validated. Similarly, Fig. 7 illustrates the training and validation loss values. The model maintains low error rates, with a validation loss of 0.42% and a training loss of 0.38%. These results highlight the model's reliability and effectiveness in accurately classifying diabetic retinopathy severity.

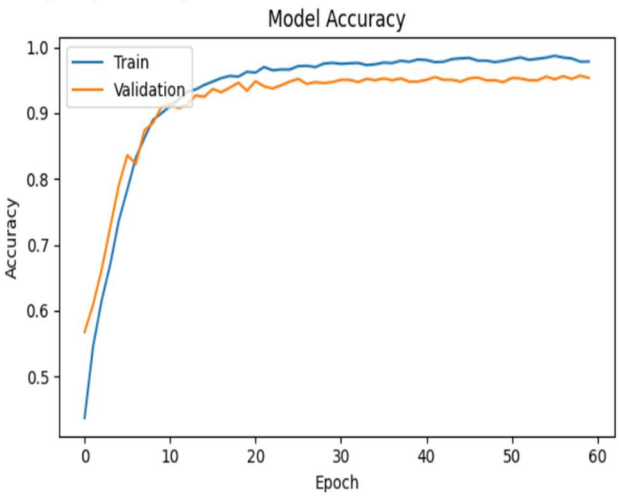

Fig. 6. Accuracy plots of the proposed model

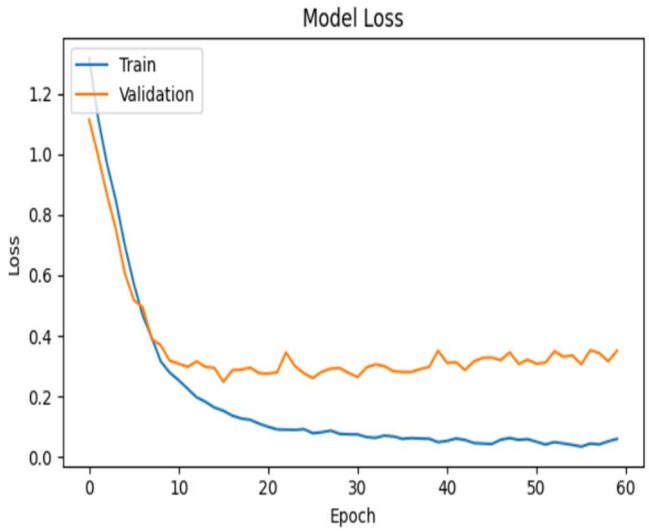

Fig. 7. The suggested model loss plots

A confusion matrix analysis provides a better method to evaluate classifier performance. The confusion matrix of our proposal is depicted in Fig. 8, showing a detailed breakdown of the number of classification results. Overall, the model demonstrates strong performance. However, when examining individual class performance, some misclassifications occur, particularly between cases of no DR and mild DR. This implies that even though the model is very accurate, there is still potential for improvement in distinguishing between cases of early-stage DR.

Table II presents the accuracy of different methods used for DR accuracy classification. The Standard Operation Procedure (SOP) achieves 75.6 % accuracy, which is relatively low. A CNN significantly improves accuracy to 93 %, demonstrating its effectiveness in feature extraction from medical images. Combining CNN with a support vector machine (SVM) further enhances accuracy to 94.7 % as SVM improves classification. The proposed model using EfficientNet achieves a promising accuracy of 96.14 %, suggesting that it optimally refines feature selection and classification. This shows how DL based approaches surpass the traditional ways of detecting DR

TABLE II. CLASSIFICATION ACCURACY OF THE PREVIOUS WORK

S.No	Method / Algorithm	Accuracy	Reference
1	EfficientNetV2	75.10	[16]
2	EfficientNetB0	79.1	[17]
3	EfficientNetB0	91.1	[18]
4	Proposed model	96.14	-

Table III displays the assessment metrics used for classification, which include precision and recall, together with the F1-score. The results indicate that our proposed method achieves consistently strong performance across all severity levels of DR. These metrics confirm that the model provides reliable and balanced classification, ensuring accurate detection in every class category.

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TABLE III. PERFORMANCE PARAMETERS

DR Class	Precision	recall	f1-score	support
No DR	0.96	0.95	0.96	306
Mild	0.95	0.96	0.95	288
Moderate	0.95	0.88	0.91	276
Sever	0.97	0.99	0.98	294
Proliferative	0.95	0.99	0.97	270

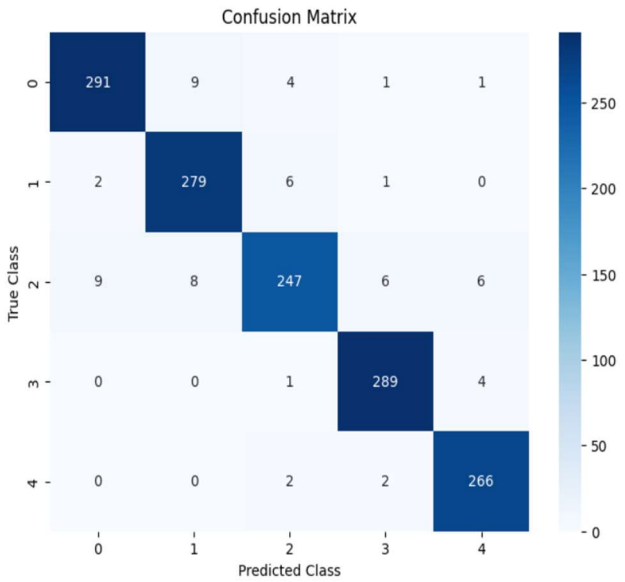

Fig. 8. Confusion matrix.

V. CONCLUSION

This paper presents an EfficientNet transfer learning performance of DR severity on retinal fundus images. By using transfer learning, the model effectively learns complex patterns even when using a relatively small sample, improves productivity, and reduces computational requirements. Data augmentation improves the model's stability as well as dataset balance. This innovation equips healthcare professionals with a robust tool for accurately evaluating the severity of DR and for fast diagnosis. The proposed model demonstrates a validation loss of approximately 0.42% and a training loss of 0.38%. Moreover, the framework exhibited improved results for precision, recall, and F1 score. In future studies, we will also concentrate on forecasts and DR segmentation, alongside categorization.

ACKNOWLEDGMENT

We thank SR University, Warangal, for providing the necessary facilities for this research work

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